llmXive Automated Review of LocateAnything: Fast and High-Quality Vision-Language Grounding with Parallel Box Decoding

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their alignment with cited evidence.

Dataset Statistics and Verification Claims There is a significant logical inconsistency regarding the dataset size in the Abstract and Section 2.4. The text states, "All 138M samples underwent automated verification; the overall rejection rate was 14.2%." If 138M is the starting number, a 14.2% rejection rate would leave approximately 118M samples, contradicting the paper's consistent use of "138M" as the final dataset size. Conversely, if 138M is the final size, the text should state that ~160M samples were generated. Furthermore, summing the query counts in Table 1 (e001) yields ~139.3M, which conflicts with the rounded "138M" figure in the text. The authors must clarify whether 138M represents the pre- or post-verification count and ensure the text matches the table summation.

Metric Discrepancies in Results In Section 3.2, the authors claim LocateAnything achieves "76.8 and 70.1 mean F1" on DocLayNet and M6Doc, respectively. However, Table 2 (e001) reports F1@mIoU scores of 77.7 for DocLayNet and 70.5 for M6Doc under the Hybrid mode. These numerical discrepancies suggest either a typo in the text or the reporting of a different metric not explicitly defined. The text must be corrected to match the precise values in the tables to ensure factual accuracy.

Unsubstantiated Baseline Throughput Section 3.2 asserts that LocateAnything is "over 10x faster than Qwen3-VL (1.1 BPS)." While the calculation (12.7 / 1.1 ≈ 11.5) is mathematically sound, the manuscript provides no table row or citation for a "Qwen3-VL" baseline with 1.1 BPS throughput. Table 3 (e001) lists "Qwen2.5-VL-3B" at 1.3 BPS, but the text specifically references Qwen3-VL. Without a cited source or a corresponding table entry for the 1.1 BPS figure, this claim lacks evidentiary support within the provided document. The authors should either include the Qwen3-VL data in the tables or provide a direct citation for the throughput figure.

Action items.

- [writing] Section 2.4 claims '138M queries' and '14.2% rejection rate' on 'All 138M samples'. This implies 138M is the pre-rejection count, yet the final dataset is stated as 138M. Clarify if 138M is pre- or post-verification to resolve the mathematical contradiction.
- [writing] Table 1 sums to ~139.3M queries, contradicting the text's '138M' claim. Additionally, Section 3.2 cites '76.8' and '70.1' F1 for DocLayNet/M6Doc, while Table 2 reports '77.7' and '70.5'. Align text values with table data.
- [writing] Section 3.2 claims '10x faster than Qwen3-VL (1.1 BPS)', but no table or citation supports the 1.1 BPS figure for Qwen3-VL. Table 3 only lists Qwen2.5-VL. Provide the source for the 1.1 BPS baseline or update the comparison table.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively illustrates the paper's versatile tasks and decoding paradigms. However, the caption contains a grammatical error with a dangling comma and a confusing note regarding the resolution of the schematic diagram.

Figure 2

Figure 2 effectively contrasts three token decoding methods visually, but the caption does not fully explain the meaning of certain symbols (scissors, warnings) or clarify what the token sequences represent, potentially confusing readers.

Figure 3

Figure 3 effectively illustrates the model architecture and the block-based output representation described in the caption. The diagram clearly maps the input processing through the Moon-ViT and Qwen2.5 components to the specific structured output blocks (Semantic, Box, Negative, End), with all labels and flow directions being legible and consistent with the text.

Figure 4

The figure illustrates the re-decoding process but contains a visual contradiction in the 'Spatial Ambiguity' panel where an erroneous token appears to be accepted before correction, conflicting with the caption's description of discarding errors. Additionally, the legend's specific role in the diagram is not clearly defined.

Figure 5

Figure 5 effectively visualizes the dataset composition with clear pie charts and a detailed legend, but the bottom of the legend is cut off, obscuring part of the 'Pointing' category breakdown, and there is a potential inconsistency between the 'unique images' count and the dataset sources listed for detection queries. issues:

- severity: writing

text:

Figure 6

The figure effectively compares box ordering strategies and decoding speeds, but the right panel is incomplete as it fails to plot the 'Parallel' throughput line defined in the legend, and the right y-axis label contains a likely typo.

Figure 7

The figure effectively demonstrates the model's robustness across diverse scenes and object counts, but it lacks a visual legend to map the box colors to the specific query categories listed in the caption, and the relationship between the bottom text labels and the 'free-form queries' is not explicitly clarified.

Figure 8

The figure effectively illustrates the data engine workflow with clear visual steps, but the caption is truncated at the end, and there are minor typos in the labels ('Categroy') and generated text ('centerx').

Figure 9

Figure 9 presents a log-scale distribution of query counts by target/box count but suffers from a mismatch between the x-axis label and caption, and unexplained missing data bars for certain task categories in higher count ranges.

Figure 10

The figure provides a visual comparison of three models on referring expression tasks, but the text labels are too small to read, and the 'LocateAnything' model exhibits a clear failure in the third row that contradicts the caption's claim of superior performance.

Figure 11

The figure effectively demonstrates the model's performance on dense objects compared to baselines, but the caption lacks explicit reference to the 'Ground Truth' column and the color coding scheme used for the different models.

Figure 12

The figure effectively demonstrates the qualitative differences in OCR performance between models, but the column headers are not explicitly defined in the caption, and the ground truth boxes appear looser than the claimed 'tight' localization of the proposed method.
Action items.

- [writing] Figure 1: The caption contains a grammatical error and likely typo: 'predicts points representing geometric units (, bounding box corners)' includes a dangling comma and missing word before the parenthesis.
- [writing] Figure 1: The caption states 'coordinate tokens in the bottom panel are rendered at high resolution to ensure legibility at print scale,' but the bottom panel is a schematic diagram, not a print-scale rendering of the actual model output.
- [science] Figure 2: The 'Generic Multi-Token Prediction' row shows a sequence of tokens (e.g., <911>, <832>) that are not explicitly defined as coordinate values in the caption, creating ambiguity about whether they represent coordinates or other data. The caption mentions 'irregular distributions' but the visual representation does not clearly illustrate this concept.
- [writing] Figure 2: The scissors icons and warning symbols lack explicit definitions in the caption or legend, making their meaning unclear to readers unfamiliar with the context.
- [science] Figure 4: The 'Spatial Ambiguity' panel shows the model discarding the erroneous block '<121>' and reverting to NTP to predict '<647>', yet the diagram visually depicts the model *accepting* the erroneous block '<121>' (highlighted with a cursor icon) before the correction arrow appears. This contradicts the caption's claim that the model 'discards the erroneous block' and misrepresents the re-decoding mechanism.
- [writing] Figure 4: The legend in the top-right corner ('Correct'/'Wrong') is not explicitly referenced in the caption or figure labels, making it ambiguous whether the red boxes represent 'Wrong' predictions or simply 'Format Irregularity'/'Spatial Ambiguity' states.
- [science] Figure 5: The bottom panel's 'Detection' category lists 'NuImages (382.1K)' and 'MOT17DET (5.3K)', but the pie chart for 'LocateAnything-Data-Queries' (138M) shows Detection at 66.9% (93M). The sum of the listed datasets (93M) matches the total, but the breakdown includes datasets like 'NuImages' which are typically video-based, raising questions about the 'unique images' count in the rightmost pie chart (12M) versus the query count.
- [writing] Figure 5: The legend for 'Pointing' (3M, 2.2%) is cut off at the bottom of the image, making it impossible to see the full list of contributing datasets (only 'Object365', 'OpenImages', 'PixmoPoints', 'RoboAfford' are visible, but the row is truncated).
- [writing] Figure 5: The legend row for 'Pointing' is partially cut off at the bottom edge of the image, truncating the list of source datasets.
- [science] Figure 5: The 'LocateAnything-Data-Images' pie chart indicates 12M unique images, yet the 'Detection' query breakdown lists datasets like NuImages (382.1K) and MOT17DET (5.3K) which are video-centric; the caption does not clarify how video frames are aggregated into 'unique images' or if the 12M count excludes these sources.
- [science] Figure 6: The right panel's legend defines 'Textual (BPS)', 'Quantized (BPS)', and

‘Parallel (BPS)’ as line plots, but the chart only displays lines for ‘Textual’ and ‘Quantized’. The ‘Parallel’ throughput line is missing, despite the caption claiming a comparison of all three methods.

- [writing] Figure 6: The right panel’s right y-axis label reads ‘Throughput−,␣BPS (Box per Second)’. The ‘−’␣symbol appears to be a typo or rendering artifact and should be removed for clarity.
- [writing] Figure 7: The caption lists four query categories (MyRedattribute, MyBluepart, MyOrangereasoning, MyGreenspatial) corresponding to the box colors, but the figure lacks a visual legend or key to map these specific names to the colors, forcing the reader to guess the mapping.
- [writing] Figure 7: The text labels below each image (e.g., ‘yellow leaves scattered among the purple leaves’) are not explicitly defined as the ‘free-form textual queries’ mentioned in the caption, creating ambiguity about whether these are the inputs or just descriptions.
- [writing] Figure 8: The caption text is truncated at the end (‘produce bo’), cutting off the description of the final output (likely ‘bounding boxes’).
- [writing] Figure 8: The top-left label contains a typo, spelling ‘Category’ as ‘Categroy’.
- [writing] Figure 8: The bottom-middle query box contains a typo, spelling ‘center’ as ‘centerx’.
- [science] Figure 9: The legend defines six task categories (Detection, GUI, Referring, OCR, Layout, Pointing), but the x-axis groups (e.g., 10-20, 20-30) show missing bars for several categories (e.g., GUI, Layout, Pointing) without explanation in the caption or visual indication (e.g., zero-height bars) of their absence.
- [writing] Figure 9: The x-axis label ‘Bbox/Point Count’ contradicts the caption’s description (‘number of targets per query’); clarify whether the axis represents targets, boxes, or points to ensure consistency.
- [science] Figure 10: The ‘LocateAnything’ column (right) shows a failure in the third row (trees) where the model produces fragmented, overlapping boxes for ‘bare trees’ instead of a single coherent region, contradicting the caption’s claim of ‘superior compositional grounding capabilities’.
- [writing] Figure 10: The text labels inside the bounding boxes (e.g., ‘black helmet with TCU logo’, ‘man wearing glasses’) are rendered at a resolution that makes them illegible in the provided image, hindering verification of the specific queries being tested.
- [writing] Figure 11: The rendered image displays a ‘Dense Object Detection’ title and a ‘Ground Truth’ column, but the caption describes it as a ‘Qualitative comparison on Dense Object Detection (DOD)’ without explicitly mentioning the inclusion of ground truth labels or the specific column layout.

- [science] Figure 11: The 'Ground Truth' column shows red bounding boxes, but the caption does not define this color coding, potentially causing confusion with the 'LocateAnything' column which uses blue boxes.
- [writing] Figure 12: The column headers 'Qwen3-VL', 'Rex-Omni', and 'LocateAnything' are not defined in the caption; the caption only mentions 'baseline models' without specifying which are which.
- [science] Figure 12: The 'Ground Truth' column displays bounding boxes that are not perfectly tight around text elements (e.g., the 'GURGEL' logo boxes include significant whitespace), which contradicts the caption's claim that LocateAnything yields 'tightly bounded boxes' compared to baselines.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized acronyms and coined terms that hinder accessibility for a broader audience. In the Introduction, the terms MTP and NTP are used immediately without definition. While "Multi-Token Prediction" and "Next-Token Prediction" are standard in the field, they must be spelled out at first use (e.g., "Multi-Token Prediction (MTP)") to comply with general readability standards. Similarly, IoU appears in the Abstract and Section 4 without expansion; "Intersection over Union" should be provided initially.

The authors introduce several proprietary or highly specific terms that function as jargon barriers. In Section 1, the phrase "Textual Digits" is used to describe a coordinate representation method. This is an unnecessary neologism; "text-based coordinates" or "tokenized coordinates" would be clearer. In Section 3.3, the concepts of "Format Irregularity" and "Spatial Ambiguity" are presented as formal failure modes. These should be rephrased into plain English, such as "syntax errors" and "unclear object boundaries," to ensure the logic is accessible without a glossary.

Furthermore, the Appendix introduces "MagiAttention" as a "Distributed framework for heterogeneous masks." Without a brief explanation of what distinguishes this from standard attention mechanisms (e.g., "a custom attention mechanism that combines causal and bidirectional masking"), the term serves only as a black-box label. Finally, the term "Atomic unit" is used repeatedly to describe bounding boxes. While conceptually useful, it is a computer science term that might be better explained as "indivisible block" or "single prediction unit" for a general scientific audience. These changes are essential to ensure the paper's contributions are understood beyond the immediate sub-field of parallel decoding. Action items.

- [writing] The manuscript relies heavily on specialized acronyms and coined terms that hinder accessibility for a broader audience. In the Introduction, the terms MTP and NTP are used immediately without definition. While "Multi-Token Prediction" and "Next-Token Prediction" are standard in the field, they must be spelled out at first use (e.g., "Multi-Token Prediction (MTP)") to comply with general readability standards. Similarly, IoU appears in the Abstract and Section 4 without expansion; "Intersection

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The manuscript presents a logical inconsistency regarding the scale of the training dataset. The Abstract states: "We curate LocateAnything-Data with more than 138 million training samples." However, Section 2.4 ("LocateAnything-Data") explicitly clarifies: "LocateAnything-Data contains 12M unique images, 138M natural language queries, and 785M annotated bounding boxes." The abstract conflates the number of *queries* with the number of *images*, creating a misleading impression of the dataset's visual diversity. This distinction is critical for evaluating the model's generalization capabilities and must be corrected to ensure the premise (large-scale diverse data) supports the conclusion (high-quality grounding).

Furthermore, the causal mechanism for the Hybrid Mode's fallback logic is insufficiently justified. Section 3.3 states: "If top-1 probability < 0.7 AND max-min difference among top-5 tokens > 80 (in [0, 1000] space), the block is discarded and NTP is used." While the thresholds are defined, the paper fails to provide a logical derivation or empirical evidence explaining why a difference of 80 units in the normalized coordinate space specifically indicates "spatial ambiguity" or "format irregularity." Without this justification, the choice of 80 appears arbitrary, weakening the logical link between the observed uncertainty and the decision to switch decoding modes.

Finally, there is a conflation of performance metrics in the narrative. The Introduction and Section 4.2 highlight a throughput of "12.7 BPS" for the Hybrid mode as the primary speed achievement. However, the ablation study (Table 3, e001) reveals that the "Fast Mode" (pure MTP) achieves 16.9 BPS. By anchoring the "2.5x speedup" claim to the Hybrid mode (12.7 BPS) rather than the theoretical maximum of the proposed architecture (16.9 BPS), the paper understates the potential of the Parallel Box Decoding method while simultaneously obscuring the cost of the fallback mechanism. The narrative should clearly distinguish between the speed of the pure parallel method and the speed of the adaptive hybrid system to accurately reflect the trade-off.
Action items.

- [science] The claim of '138M unique images' in the Abstract contradicts Section 2.4

(LocateAnything-Data) which states '12M unique images'. This internal inconsistency regarding the dataset scale must be resolved.

- [writing] The fallback mechanism in Section 3.3 specifies a threshold of 'max-min difference > 80' in [0, 1000] space, but the text does not explain the causal link between this specific value and 'unreliability' or 'format violations'. The logic for this heuristic is missing.
- [writing] Table 1 (e000) reports LocateAnything-3B throughput as 12.7 BPS, while Table 2 (e001) and the text in Section 4.2 report Hybrid Mode throughput as 12.7 BPS but Fast Mode as 16.9 BPS (Table 3). The main text claims '12.7 BPS (Hybrid)' is 'over 10x faster than Qwen3-VL (1.1 BPS)', but the ablation table shows PBD (Fast) at 16.9 BPS. The paper conflates the 'Hybrid' throughput with the 'Fast' potential in the narrative, obscuring the actual speed gain of the proposed method.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that extend beyond the immediate evidence provided in the text and tables, particularly regarding the magnitude of speed improvements, the statistical validity of dataset quality assertions, and the definition of "State-of-the-Art" performance.

First, the Introduction claims the method achieves "up to 2.5x higher decoding throughput." While Table 1 shows a 2.54x improvement over Rex-Omni (12.7 vs 5.0 BPS), the same table shows an 11.5x improvement over Qwen3-VL (12.7 vs 1.1 BPS). By selecting the 2.5x figure, the authors appear to be minimizing the reported speed advantage against the most relevant modern baseline (Qwen3-VL) while highlighting a comparison against a potentially weaker or older baseline (Rex-Omni). This selective reporting overstates the "frontier" advancement by obscuring the full range of speedups. The claim should be revised to "up to 11.5x" or explicitly qualified as "2.5x over Rex-Omni."

Second, the Abstract asserts that "All 138M samples underwent automated verification" and cites a "99.4% agreement" from a "500-sample spot-check." This is a statistical overreach. A sample size of 500 out of 138,000,000 (0.00036%) is insufficient to support a definitive claim about the quality of the *entire* dataset with such high precision. The 99.4% figure likely represents the agreement rate *within the spot-check*, not a verified quality metric for the full 138M. The phrasing "All... underwent verification" implies a deterministic check for every sample, which contradicts the reliance on a tiny spot-check for the aggregate quality metric. The authors must clarify that the 99.4% is an estimate with a specific confidence interval or rephrase to avoid implying total verification.

Third, the "High-Quality" claim relies on the Hybrid Mode achieving a "SOTA mean F1 of 60.3" on ScreenSpot-Pro. However, Table 2 in the Additional Experiments section explicitly shows that the Slow Mode achieves 60.5 F1, which is higher than the Hybrid mode's 60.3. If the Slow Mode is the true SOTA, the text should claim "SOTA performance achieved by the Slow Mode" rather than attributing the SOTA status to the Hybrid mode, which is slightly lower. This distinction is vital for the paper's narrative of balancing speed and accuracy; if the "fast" mode is not the SOTA, the "High-Quality" claim for the Hybrid mode is weakened.

Finally, the description of the Hybrid Mode's fallback mechanism lacks empirical validation. The paper claims the fallback (triggered by specific probability thresholds) effectively handles "Format Irregularity" and "Spatial Ambiguity" without quantifying how often this fallback is actually invoked. Without data on the fallback frequency (e.g., "fallback occurred in 5% of cases"), the claim that the mode "preserves speed gains" is an assumption rather than a demonstrated fact. The paper overreaches by presenting the mechanism as a solved problem without showing the trade-off statistics.

Action items.

- [science] The paper makes several claims that extend beyond the immediate evidence provided in the text and tables, particularly regarding the magnitude of speed improvements, the statistical validity of dataset quality assertions, and the definition of "State-of-the-Art" performance. First, the Introduction claims the method achieves "up to 2.5x higher decoding throughput." While Table 1 shows a 2.54x improvement over Rex-Omni (12.7 vs 5.0 BPS), the same table shows an 11.5x improvement over Qwen3-VL (12

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a significant technical contribution to vision-language grounding but requires clarification on data safety and ethical sourcing before acceptance.

Data Provenance and Synthetic Bias: The core of the training data, "LocateAnything-Data," is largely synthetic, generated by Qwen3-VL and SAM 3 (Appendix: "LocateAnything-Data Construction"). While the authors report a 99.4% agreement rate on a 500-sample spot-check, this sample size is statistically insufficient to guarantee the absence of systemic bias or safety violations across 138M samples. The reliance on teacher models (Qwen3-VL) to generate queries and boxes risks propagating the biases, stereotypes, or hallucinations inherent in those foundation models. The paper should explicitly discuss the strategy for mitigating "model collapse" or bias amplification in this synthetic pipeline.

Negative Sample Construction: The inclusion of 22M negative samples is a strong safety feature to reduce hallucination. However, the method of construction is vague ("Explicitly constructed"). If these negatives were generated by adversarially perturbing images or synthesizing harmful scenarios to teach the model what *not* to detect, the authors must detail the safety guardrails used to ensure the training process itself does not learn to associate harmful concepts with specific visual patterns.

Content Safety and Toxicity: The training data aggregates numerous public datasets, including "Hateful Memes" and "Chinese-Meme" (Supp Table e002). While these are standard benchmarks, the paper lacks a dedicated section on content moderation. Given the model's ability to ground text in images, there is a risk it could be used to locate or identify sensitive content (e.g., hate speech in images, private individuals). The authors must confirm that a toxicity filter was applied to the text queries and that the dataset does not contain PII or non-consensual imagery, particularly from web-scraped sources like Unsplash.

Privacy and Consent: The use of 12M unique images from diverse sources (Unsplash, SA-1B, etc.) raises standard data privacy concerns. The authors should include a statement confirming compliance with the licenses of these datasets and describing any steps taken to redact or exclude images containing faces or sensitive personal information, especially given the high-resolution nature of the grounding task.

These issues are primarily related to the *construction* and *curation* of the dataset rather than the model architecture itself. Addressing them will ensure the model is deployed responsibly.

Action items.

- [writing] The manuscript presents a significant technical contribution to vision-language grounding but requires clarification on data safety and ethical sourcing before acceptance. Data Provenance and Synthetic Bias: The core of the training data, "LocateAnything-Data," is largely synthetic, generated by Qwen3-VL and SAM 3 (Appendix: "LocateAnything-Data Construction"). While the authors report a 99.4% agreement rate on a 500-sample spot-check, this sample size is statistically insufficient to guarantee

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence supporting the central claims of LocateAnything is currently insufficient due to missing statistical context and potential confounding variables in the experimental setup.

First, the data quality claims in the Abstract are not statistically robust. The authors state that "a 500-sample spot-check revealed 99.4% agreement with teacher labels." For a sample size of 500, the 95% confidence interval for a 99.4% success rate is approximately [98.1%, 99.9%]. While high, the paper fails to report this interval or the specific criteria used for the "agreement" metric (e.g., IoU threshold for box matching). Furthermore, the claim of a "14.2% rejection rate" for 138M samples implies a massive automated filtering process, yet the methodology for this automated verification is not detailed, leaving open the possibility of systematic bias in the remaining data.

Second, the throughput comparisons are vulnerable to confounding factors. The paper claims LocateAnything is "10x faster than Qwen3-VL (1.1 BPS)" and "2.5x faster than Rex-Omni (5.0 BPS)." However, the "Experimental Setup" section (Supp) only specifies that throughput was evaluated on COCO, without explicitly confirming that all baselines were run on identical hardware (GPU model, count, memory bandwidth) or with identical software optimizations (e.g., FlashAttention, quantization levels). If the baselines were run in a less optimized environment or on older hardware, the speedup claim is invalid. The authors must provide a hardware/software matrix for all reported throughput numbers.

Third, the ablation study (Table 3) presents a trade-off that requires deeper analysis. While PBD (Fast) achieves 16.9 BPS, its F1 score (49.6) is significantly lower than PBD (Slow) (52.1). The Hybrid mode (13.2 BPS, 51.6 F1) is claimed to be the "optimal choice," but the paper does not report the fallback rate (i.e., how often the model had to switch from Fast to Slow mode). If the fallback rate is high, the effective throughput would be much lower than 13.2 BPS, undermining the efficiency claim.

Finally, there is a logical inconsistency in the dataset statistics (Table 2). The table lists "Training queries" as 138M and "Validation queries" as 138M, with a "Total queries" of 138M. This implies the validation set is the same size as the training set, or the total is misreported. Given the scale of the dataset, this ambiguity makes it difficult to assess the true volume of data used for training versus evaluation.

To proceed, the authors must provide confidence intervals for data quality metrics, a detailed hardware/software comparison for speed benchmarks, fallback rate statistics for the Hybrid mode, and a corrected dataset split table.

Action items.

- [science] The claim of 138M training samples with a 14.2% rejection rate and 99.4% spot-check agreement (Abstract) lacks statistical rigor. The paper must report the confidence interval for the 99.4% agreement metric (n=500) and clarify the rejection criteria. Without this, the data quality claim is unsubstantiated.
- [science] The throughput comparison (12.7 BPS vs 1.1 BPS) is potentially confounded by hardware and implementation differences. The paper must explicitly state the hardware (GPU model, count, memory) and software stack (framework version, precision) used for all baselines to ensure a fair speed comparison.
- [science] The ablation study (Table 3) shows PBD (Fast) has lower F1 (49.6) than PBD (Slow) (52.1) but claims a 2.5x speedup. The paper must provide a detailed breakdown of the fallback frequency in Hybrid mode and the specific cost of the fallback mechanism to validate the "optimal" trade-off claim.
- [science] The dataset statistics table (Table 2) lists 138M queries but the validation set is also listed as 138M queries, which is logically inconsistent for a standard train/val split.

The authors must clarify the data split strategy and correct the reported numbers to avoid confusion about the actual training volume.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The manuscript presents a novel Parallel Box Decoding (PBD) approach but lacks rigorous statistical validation for its core empirical claims.

First, the data quality assertions in the Abstract are statistically unsupported. The claim of a "14.2% rejection rate" across 138M samples and "99.4% agreement" on a 500-sample spot-check lacks necessary statistical context. The authors must report the confidence interval for the 99.4% agreement rate (which, for n=500, is approximately ±1.3% at 95% confidence) and explicitly define the sampling strategy used for the spot-check. Without this, the generalizability of the quality control to the full 138M dataset is unproven.

Second, the throughput and performance comparisons rely solely on point estimates. For instance, the claim of "12.7 BPS" versus "1.1 BPS" (Section 5.2) is presented without standard deviations, confidence intervals, or results from statistical significance tests (e.g., paired t-tests). In systems benchmarking, variance due to hardware thermal throttling or batch size fluctuations is common; a single measurement is insufficient to substantiate a "10x faster" claim.

Third, the ablation studies (Table 3) involve multiple pairwise comparisons of F1 scores across different modes and representations. The manuscript fails to address the multiple-comparisons problem. Without corrections (e.g., Bonferroni), the probability of observing at least one statistically significant difference by chance increases with the number of tests performed. The authors should either apply such corrections or provide p-values for the reported improvements.

Finally, all performance metrics in Tables 2, 4, and 5 are reported as single values. Standard practice in computer vision requires reporting standard errors or 95% confidence intervals to distinguish genuine model improvements from evaluation noise. The absence of these metrics makes it impossible to assess the statistical robustness of the reported "SOTA" results.

Action items.

- [science] The manuscript presents a novel Parallel Box Decoding (PBD) approach but lacks rigorous statistical validation for its core empirical claims. First, the data quality assertions in the Abstract are statistically unsupported. The claim of a "14.2% rejection rate" across 138M samples and "99.4% agreement" on a 500-sample spot-check lacks necessary statistical context. The authors must report the confidence interval for the 99.4% agreement rate (which, for n=500, is approximately ±1.3% at 95% confid

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a compelling technical contribution, but the writing quality requires minor revisions to ensure clarity, consistency, and professional polish.

First, there is a noticeable typo in the caption of Figure 2 (e001), where "categroy-per-query" is misspelled as "categroy". This error propagates to the label reference `\label{fig:categroy-per-query}` and the text referencing it. This must be corrected to "category" to maintain professional standards.

Second, there is a significant inconsistency regarding dataset statistics that confuses the reader. The Abstract and Section 2.4 ("LocateAnything-Data") state the dataset contains "138M natural language queries." However, Table 2 (e001) lists "Training queries: 138M" and "Validation queries: 138M," which implies a total of 276M queries or suggests a duplication error in the table. The text in Section 2.4 also mentions "138M unique images" and "785M annotated bounding boxes," but the query count ambiguity remains. The authors must clarify whether the 138M figure represents the total dataset size or just the training split, and ensure the tables align with the text.

Third, the flow of the "On-Demand Inference Modes" section (Section 2.3) is slightly disjointed. The transition between the definition of the fallback mechanism (probability thresholds) and the enumeration of the three modes could be smoother. Additionally, the phrase "Format Irregularity (malformed syntax at category boundaries)" in Section 2.3 is slightly vague; specifying that this refers to "token-level syntax errors at block boundaries" would improve precision.

Finally, while the tables are generally well-structured, Table 3 (e001) contains a minor grammatical awkwardness in its caption: "The results of the *Hybrid Mode* are reported here." Changing "here" to "herein" or rephrasing to "This table reports the results..." would improve the academic tone. Similarly, the capitalization of "Boxes Per Second" varies between the text (Section 3.2) and table headers; standardizing this to "Boxes Per Second (BPS)" on first use and "BPS" thereafter is recommended.

Addressing these specific points will significantly enhance the readability and credibility of the manuscript.

Action items.

- [writing] Correct the typo 'categroy' to 'category' in the caption of Figure 2 (e001) and the corresponding label reference.
- [writing] Resolve the inconsistency in dataset statistics: The abstract and Section 2.4 state 138M queries, while Table 1 (e001) lists 138M queries but Table 2 (e001) lists 138M for both training and validation, implying a total of 276M or a duplication error. Clarify the exact split.

- [writing] Fix the grammatical error in Table 3 (e001) caption: 'The results of the *Hybrid Mode* are reported here' should be 'The results of the *Hybrid Mode* are reported herein' or rephrased for better flow.
- [writing] Standardize the capitalization of 'Boxes Per Second' in the text (e.g., Section 3.2) versus 'BPS' in table headers to ensure consistent terminology throughout the manuscript.